

DevOps Documentation

Program 8: Create Kubernetes Pods using YAML Descriptors

Program-8

—>nodejs-configmap.yaml

—>nodejs-pod.yaml

—>[server.js](#)

S1. Write a [server.js](#) app

```
JS server.js  X
JS server.js > ...
~/DevOpslab/program8/server.js "http");
2  const port=3000;
3  const server = http.createServer((req,res)=>{
4    res.end("Hello from Node.js running inside Kubernetes!");
5  });
6
7  server.listen(port,()=>{
8    console.log(`Server running at port ${port}`);
9  });
```

S2. Create a ConfigMap for the code

```
! nodejs-configmap.yaml  X
! nodejs-configmap.yaml
1  apiVersion: v1
2  kind: ConfigMap
3  metadata:
4    name: nodejs-app-config
5  data:
6    server.js: |
7      const http = require("http");
8      const port = 3000;
9
10     const server = http.createServer((req, res) => {
11       res.end("Hello from Node.js running inside Kubernetes!");
12     });
13
14     server.listen(port, () => {
15       console.log(`Server running at port ${port}`);
16     });
17
```

S3. Create a Pod YAML using [Node.js](#) as a base image

```
! nodejs-pod.yaml x
! nodejs-pod.yaml
1  apiVersion: v1
2  kind: Pod
3  metadata:
4    name: program-8
5    labels:
6      app: nodejs
7  spec:
8    containers:
9      - name: nodejs
10       image: node:18
11       command: ["node", "/usr/src/app/server.js"]
12       volumeMounts:
13         - name: app-code
14           mountPath: /usr/src/app
15       ports:
16         - containerPort: 3000
17     volumes:
18       - name: app-code
19         configMap:
20           name: nodejs-app-config
```

S4. Apply the YAML file

kubectl apply -f nodejs-configmap.yaml

Kubectl apply -f nodejs-pod.yaml

```
• 1rv24mc113_varun@varun-srivastava:~/DevOpslab/program8$ kubectl apply -f nodejs-configmap.yaml
configmap/nodejs-app-config created
• 1rv24mc113_varun@varun-srivastava:~/DevOpslab/program8$ kubectl apply -f nodejs-pod.yaml
pod/program-8 created
○ 1rv24mc113_varun@varun-srivastava:~/DevOpslab/program8$
```

S5. Check pod status

kubectl get pods

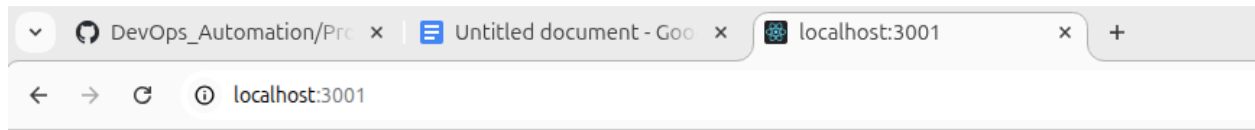
```
• 1rv24mc113_varun@varun-srivastava:~/DevOpslab/program8$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-8nvjh   1/1     Running   1 (3h6m ago)  4h20m
nginx-deployment-6f9664446b-xkg4x   1/1     Running   1 (3h6m ago)  4h20m
nginx-deployment-6f9664446b-z9m55   1/1     Running   1 (3h6m ago)  4h20m
program-8                            1/1     Running   0           107s
○ 1rv24mc113_varun@varun-srivastava:~/DevOpslab/program8$
```

S6. Access the [Node.js](#) app

kubectl port-forward pod/nodejs-pod 3001:3000

```
❖ 1rv24mc113_varun@varun-srivastava:~/DevOpslab/program8$ kubectl port-forward pod/program-8 3001:3000
Forwarding from 127.0.0.1:3001 -> 3000
Forwarding from [::1]:3001 -> 3000
```

S7. Check the output in <http://localhost:3000>



Hello from Node.js running inside Kubernetes!